Bathymetry as a Coupled Radiative Transfer Inverse Problem: Rationale, Physics and Implementation Specification for *photic*

Design specification

16 October 2026

Abstract

photic inverts atmospherically corrected reflectance with Lee's semi-analytical model, one pixel neighbourhood at a time. ACOLITE does the atmospheric correction beforehand, for each scene separately. This document specifies a replacement for both steps, built in stages.

The forward model becomes the radiative transfer equation (RTE), the integro-differential transport equation that HydroLight solves. It covers the atmosphere, the sea surface, the water column and the bottom, and it is evaluated over each band's spectral response. The RTE is never solved inside the fit. Its exact solution is precomputed as a lookup table over a few dimensionless variables, using a discrete-ordinate solver written for *photic*. The fit interpolates the table, which gives the speed of a table with an interpolation error that is measured and kept below the sensor noise. The atmosphere is handled the same way, from the 6SV-based tables that ACOLITE distributes.

The inversion keeps *photic*'s structure: a bounded Levenberg–Marquardt fit over a pixel window and all scenes at once, with depth and bottom shared across scenes and water properties per scene. In the later stages each scene's aerosol joins the fitted parameters. Part I gives the rationale and the physics. Part II is the specification: modules, data formats, commands, numerical methods, milestones, tests and acceptance criteria.

Contents

I Rationale and physics **2**

1 Does this already exist? **3**

2 Why change the forward model **3**

3 The forward problem **4**
3.1 The coupled RTE 4
3.2 Water IOPs 4
3.3 The measurement 5
3.4 A hierarchy of approximations 5

4 Lookup tables from the PDE solution **5**
4.1 Why the table is small 5
4.2 Illumination 6

5 The inverse problem **6**
5.1 Unknowns . . . 6
5.2 Objective . . . 6
5.3 Gradients . . . 7
5.4 The bottom mixture and discrete choices . . . 7

II Implementation specification **7**

6 Scope and milestones **7**

7 Software architecture **8**

8 The water-column solver (`rtwater.c`) **8**
8.1 Problem . . . 8
8.2 Discretisation . . . 8
8.3 API . . . 9
8.4 Cost . . . 10

9 Tables (`rttable.c`) **10**
9.1 Axes . . . 10
9.2 Build . . . 10
9.3 Interpolation . . . 11
9.4 Validation of a table . . . 11
9.5 Derivatives for the fit . . . 11

10 Changes to `samodel.c` **11**

11 The atmosphere (`atmos.c`, M3) **12**
11.1 Data . . . 12
11.2 TOA model . . . 12
11.3 In the fit . . . 13
11.4 Inputs . . . 13

12 Commands and data **13**
12.1 Commands . . . 13
12.2 New data files . . . 13

13 Tests **14**
13.1 Unit and solver tests (`model/tests/rt_check.c`, `make test-rt`) . . . 14
13.2 Benchmark . . . 14

14 Risks and open questions **15**

A A global table (optional) **15**

B Global solve with the adjoint (M6, optional) **15**

Part I
Rationale and physics

1 Does this already exist?

We found no published method that does all of the following at once:

1. solves the full RTE, not a parametric approximation, for the coupled atmosphere–water–bottom system;
2. retrieves bottom depth;
3. inverts several scenes jointly, with depth and bottom shared across them.

Each part exists separately.

- **Exact water-column RT.** HydroLight solves the RTE in the water column by invariant imbedding, for given inherent optical properties (IOPs), depth and bottom reflectance [10, 13]. OSOAA [3] and CAO-DISORT [9] solve the coupled atmosphere–ocean problem. All three are forward models.
- **Coupled atmosphere–ocean inversion.** Joint retrievals of aerosol and water properties exist for open and coastal ocean colour. Examples are the multi-angle polarimetric retrieval developed for PACE [7] and POLYMER [15]. None models a bottom.
- **Bayesian shallow-water inversion.** Thompson et al. [17] solve for the most probable depth, seafloor reflectance and water properties. Their atmospheric correction is a separate step run beforehand [18].
- **Lookup tables of RT solutions.** Spectrum matching against tables of HydroLight spectra over discrete water, bottom and depth combinations is established [11].
- **Multi-temporal bathymetry.** Published multi-image satellite methods mostly composite the images, then fit an empirical or machine-learning model. *photic*'s temporal extension [1] is the joint physical case, but it uses Lee's model on corrected reflectance.

2 Why change the forward model

Validation of *photic* found the following (`validation/EXPERIMENTS.md`; `REVIEW.md`, §2d):

1. **False depths fit as well as true ones.** With one bottom type selected by fit, deep water at the Gulf entrance gets false shallow depths: RMSE 10.8 m, bias −6.8 m. Their misfit is as low as that of the correct solution.
2. **Errors the fit cannot see.** The tables were sampled at band centres. Averaged over the Sentinel-2 B2 response, pure-water absorption is 32% higher. In the synthetic tests that gives depth errors of −17% to +28% at a reduced $\chi \leq 0.15$.
3. **The correction is fixed before the inversion.** Changing ACOLITE's aerosol retrieval (tiled, or with adjacency correction) moves tier 4 by more than the noise floor, in opposite directions at different sites. The inversion cannot push back on an aerosol estimate that its own fit rejects.

4. **Lee's model approximates the RTE.** It is a fit to HydroLight runs. With four visible bands, a structural error of a few per cent becomes metres of depth.

Joint processing of several scenes holds constraints that per-scene correction throws away. The seabed is the same in every scene up to tide, so an aerosol estimate that makes one scene's bottom signal consistent with the others is more likely to be right.

3 The forward problem

3.1 The coupled RTE

$\mathbf{x}$ is horizontal position and z height (positive upwards, mean sea surface at $z = 0$). Scene t has tide η_t, so its water column is $-D_t < z < 0$ with

$$D_t(\mathbf{x}) = H(\mathbf{x}) + \eta_t(\mathbf{x}), \tag{3.1}$$

where H is the depth below the common datum (*photic*'s `HTIDE` convention). For each wavelength λ and direction $\boldsymbol{\xi}$ ($\mu = \boldsymbol{\xi} \cdot \hat{\mathbf{z}}$), the radiance satisfies

$$\boldsymbol{\xi} \cdot \nabla L + c\, L = \int_{S^2} \beta(\boldsymbol{\xi}' \to \boldsymbol{\xi})\, L(\boldsymbol{\xi}')\, \mathrm{d}\Omega', \tag{3.2}$$

in the atmosphere ($0 < z < z_T$) and in the water. $c = a + b$ is the beam attenuation and β the volume scattering function. The boundary conditions are:

$$z = z_T,\ \mu < 0: \qquad L = F_0\, \delta(\boldsymbol{\xi} - \boldsymbol{\xi}_0), \tag{3.3}$$

$$z = 0: \qquad \begin{pmatrix} L^{\uparrow}(0^+) \\ L^{\downarrow}(0^-) \end{pmatrix} = \begin{pmatrix} \mathcal{R}_{aw} & \mathcal{T}_{wa} \\ \mathcal{T}_{aw} & \mathcal{R}_{wa} \end{pmatrix} \begin{pmatrix} L^{\downarrow}(0^+) \\ L^{\uparrow}(0^-) \end{pmatrix}, \tag{3.4}$$

$$z = -D_t,\ \mu > 0: \qquad L = \frac{\rho_b}{\pi} \int_{\mu' < 0} |\mu'|\, L(\boldsymbol{\xi}')\, \mathrm{d}\Omega'. \tag{3.5}$$

The surface operators are Fresnel reflection and transmission, averaged over the Cox–Munk distribution of wave slopes [4] for a rough surface. The bottom is Lambertian, with a mixture of the library's types:

$$\rho_b(\mathbf{x}, \lambda) = \sum_k q_k(\mathbf{x})\, B_k(\mathbf{x})\, \hat{\rho}_k(\lambda), \qquad q_k \geq 0,\ \sum_k q_k = 1. \tag{3.6}$$

Here $\hat{\rho}_k$ are the library's shapes, q_k the cover fractions, and B_k each type's brightness within the library's bounds. This is *photic*'s existing `BOTTOM_BOUNDS` parameterisation. Real seabeds are mixed within a pixel, so fitting a single type per pixel is much worse against measured bottoms (§14).

3.2 Water IOPs

These are *photic*'s present IOP model, with the scattering coefficient and phase function made explicit:

$$a = a_w + (a_0 + a_1 \ln P)_+\, P + G\, e^{-S_g(\lambda - 440)}, \tag{3.7}$$

$$b_{bp} = X\, (440/\lambda)^{Y}, \qquad b_p = b_{bp}/B_p, \qquad b_w = 2\, b_{bw}, \tag{3.8}$$

$$\beta = b_w\, \tilde{\beta}_w + b_p\, \tilde{\beta}_{\mathrm{FF}}(\,\cdot\,; B_p). \tag{3.9}$$

- $\tilde{\beta}_w(\psi) \propto 1 + 0.835 \cos^2 \psi$ is the pure-water phase function.

- $\tilde{\beta}_{\mathrm{FF}}$ is the Fournier–Forand phase function [6] with backscatter fraction B_p, fixed by default at 0.0183, the Petzold average.
- The tables a_w, b_{bw}, a_0 and a_1 come from `model/data`.
- $(\cdot)_+$ is the existing clamp at zero.

3.3 The measurement

Band j of scene t is the response-weighted average

$$R_{rs,j} = \frac{\int S_j(\lambda)\, F_0(\lambda)\, R_{rs}(\lambda)\, \mathrm{d}\lambda}{\int S_j(\lambda)\, F_0(\lambda)\, \mathrm{d}\lambda} \tag{3.10}$$

for water-leaving reflectance (stages M1 and M2). At top of atmosphere (stage M3) the same average is applied to ρ^{TOA}. S_j comes from `model/data/srf`; F_0 is the extraterrestrial solar irradiance (new data file, §12).

3.4 A hierarchy of approximations

Column (levels 0–1). The water and bottom are a plane-parallel column at each pixel, exact for horizontally uniform water and bottom. The atmosphere is plane-parallel. At level 1, neighbouring pixels are coupled through the atmospheric adjacency kernel [16, 20].

3D (level 2). Monte Carlo or SHDOM [5], used only to check the column approximation at reef edges. It is never used inside the fit.

4 Lookup tables from the PDE solution

The RTE is solved offline and stored as a table that the fit interpolates. The table's only error is interpolation, which can be measured and reduced. The structural error of Lee's fitted model cannot.

4.1 Why the table is small

1. **Similarity.** For a homogeneous column, the monochromatic RTE depends on the IOPs and depth only through three quantities:

$$\omega = \frac{b}{c}, \qquad \tau = c\, D, \qquad f_w = \frac{b_w}{b}. \tag{4.1}$$

   ω is the single-scattering albedo and τ the optical depth. f_w is the share of molecular scattering, which sets the mixed phase function (3.9) for a fixed B_p. Wavelength is not a dimension: (3.7)–(3.8) map each wavelength to its own (ω, τ, f_w). So the same table serves every band and every response sub-sample. The interface depends on wavelength only through the refractive index of water, which is held at 1.34.

2. **The bottom in closed form.** For a Lambertian bottom, the remote-sensing reflectance just above the surface, in any direction, is exactly

$$R_{rs}(\rho_b) = R_0 + \frac{A\, \rho_b}{1 - C\, \rho_b}. \tag{4.2}$$

- R_0 is the reflectance over a black bottom.
- $A = E_{b0}\,L_e$, where E_{b0} is the irradiance reaching a black bottom and L_e the radiance leaving the surface per unit irradiance emitted isotropically by the bottom.
- C is the fraction of that emitted irradiance which the column and surface return to the bottom.

None of these depends on ρ_b. Two solutions with the same operator determine them: one over a black bottom, and one with an isotropically emitting bottom and no sky.

3. **One solve gives every direction.** A discrete-ordinate solution gives the radiance in all directions, so view angle and azimuth add storage but no solves.

4.2 Illumination

The table is built for two illuminations of the surface: the direct sun at θ_s, and a uniform sky of unit irradiance. A scene band's reflectance is

$$R_{rs} = (1 - s)\,R_{rs}^{\mathrm{dir}} + s\,R_{rs}^{\mathrm{sky}}, \tag{4.3}$$

where s is the diffuse fraction of the downwelling irradiance. s comes from the atmosphere tables at the scene's aerosol optical depth, or from a Rayleigh-only estimate when none is given. The uniform sky is an approximation. Its effect on near-nadir R_{rs} is small and is measured in test T6 (§13).

5 The inverse problem

5.1 Unknowns

Symbol	Meaning	Shared across scenes?	Stage
H	depth below datum, per region of the window	yes	M1
q_k, B_k	cover fractions and brightnesses per region	yes	M1
$\ln P, \ln G, \ln X$	water IOPs per scene (window-constant)	no	M1
Δ_t	glint offset per scene (`GLINT_OFFSET`)	no	M1
$\tau_{a,t}$	aerosol optical depth at 550 nm per scene	no	M3
m_t	aerosol model per scene (discrete)	no	M3
$\delta\eta_t$	tide-datum offset per scene	no	M4

5.2 Objective

For each window (*photic*'s pixel neighbourhood), the fit minimises

$$\chi^2 = \sum_{t,j,r} \frac{\left(R_{t,j}^{\mathrm{model}}(\theta) - R_{t,j}^{\mathrm{obs}}\right)_r^2}{\sigma_{t,j}^2} + \sum_i \frac{(\theta_i - \theta_i^{\mathrm{prior}})^2}{\sigma_i^2}, \tag{5.1}$$

by bounded Levenberg–Marquardt, as `samodel_lm` does now. The prior rows are the existing continuity, coherence and depth-prior rows, plus the aerosol prior of stage M3. At stage M3, R is the TOA reflectance and the bands include NIR and SWIR.

5.3 Gradients

At stages M1–M4 the forward model is local to the window, so the Jacobian follows from the chain rule through the table interpolant (§9.5). No adjoint is needed. The adjoint RTE becomes necessary only for the optional global solve (M6), in which the atmosphere and adjacency couple all pixels. The design for that is in Appendix B.

5.4 The bottom mixture and discrete choices

The bottom is a mixture (3.6), fitted with the other continuous parameters as *photic* does now (`ALL_BOTTOMS 1`). A mixture is still a Lambertian bottom, so the closed form (4.2) holds with $\rho_b(\lambda)$ the mixture's reflectance in each sub-sample, and the table needs no change.

The aerosol model is a discrete choice. It is made by fitting each candidate and keeping the lowest χ^2. The optically deep alternative is the existing test, with $R_0(\tau \to \infty)$ from the table.

Part II
Implementation specification

6 Scope and milestones

Each milestone is a separate branch and change. Its forward model is switchable, and the existing model stays the default until the acceptance rule of `validation/README.md` (tier 3 and tier 4 depth metrics improve, with no regressions) says otherwise.

	Deliverable	Done when
M0	Band averaging of the existing model (`SCENE ... SRF`); spectral data in files.	Done on branch `data-inputs`; benchmarked; adopted as the acceptance rule decides.
M1	`rtwater.c`: discrete-ordinate solver for the water column.	Passes tests T1–T4 (§13).
M2	`rttable.c`: per-scene tables, interpolation and derivatives. RT forward model in `samodel.c` behind `SET FORWARD RT`; ACOLITE R_{rs} input.	T5–T8 pass; `make test-rt` passes; tier 2 RT-vs-OSOAA error $< 1\%$; tier 3 and 4 benchmarked against M0.
M3	`atmos.c`: reader for ACOLITE's LUTs; TOA forward model; aerosol per scene in the fit; NIR/SWIR bands.	T9–T11 pass; tier 4 run from L1 TOA reflectance, compared with ACOLITE input.
M4	Tide-datum offsets per scene; optically deep test from the table.	Tier 4, including Gulf entrance false depths, benchmarked.
M5	Adjacency (level 1).	Reef-edge 3D check (level 2); tier 4.
M6	Optional global solve with the adjoint (Appendix B).	Research; not scheduled.

7 Software architecture

New modules follow the conventions in `CLAUDE.md`: C17, 2-space indentation in new files, `//` comments in sentence case, British spelling, globals declared in `common.h` and defined in `common.c`. OpenMP is not used.

File	Responsibility
`rtwater.c/.h`	Discrete-ordinate solution of one homogeneous water layer under a Fresnel interface, over a black or isotropically emitting Lambertian bottom (§8). Pure numerics, with no *photic* state.
`rttable.c/.h`	Table axes, build, cache file I/O, validation against direct solves, and interpolation with derivatives (§9).
`atmos.c/.h` (M3)	ACOLITE LUT reader; band-averaged path reflectance, transmittances, spherical albedo and diffuse fraction as functions of τ_a, with derivatives (§11).
`spectra.c`	Gains the solar irradiance F_0 and the band sub-sampling used by (3.10).
`samodel.c`	A forward-model switch in `samodel_Rrs` and the Jacobian code (§10); aerosol parameters (M3).
`bam.c`	Commands and `SCENE` keywords (§12).
`model/tests/`	`rt_check.c` (T1–T8), and `synth_gen` in RT mode for `make test-rt`.
`validation/`	`tier2_osoaa.py` gains an RT-mode comparison; `rt_table_check.py` compares tables with OSOAA (T4).

Dependencies. `rtwater.c` needs a real symmetric or nonsymmetric eigensolver for matrices up to 64×64 and an LU solver. Write them in C (Householder reduction and QR iteration, as in DISORT's `ASYMTX` [14]), or use LAPACK if the build adds it. The default is in-tree code, so that the dependencies stay NetCDF and PGPLOT.

8 The water-column solver (`rtwater.c`)

8.1 Problem

The solver handles a homogeneous slab of optical thickness τ and single-scattering albedo ω, with a phase function given by its Legendre moments χ_ℓ, $\ell = 0..2N-1$. It lies under an interface with relative refractive index $n_w = 1.34$, on a Lambertian bottom. It returns, per unit downwelling irradiance just above the surface:

- the remote-sensing reflectance just above the surface at the requested view direction;
- the irradiance reaching the bottom;

for the three source cases: direct sun, uniform sky, and emitting bottom with no sky.

8.2 Discretisation

- **Streams.** Use N streams per hemisphere in the water (default $N = 24$, at most 32). They are split into a refracted cone (0 to the critical angle, $\arcsin(1/n_w) \approx 48.3°$) and a total-internal-reflection region, each with its own double-Gauss quadrature, as CAO-DISORT does [9]. This keeps the interface's discontinuity at the critical angle on a quadrature boundary.

- **Azimuth.** Expand in Fourier modes $m = 0..M$, with $M = 2N - 1$. The Lambertian bottom couples only $m = 0$. The direct beam drives every mode. The sky drives only $m = 0$.
- **Phase function.** The Fournier–Forand moments are computed once per B_p by Gauss–Legendre integration of the analytic phase function, with 2000 or more nodes concentrated at small angles. Use δ-M truncation [21], and correct the single scattering into the view direction with the full phase function, as in Nakajima and Tanaka's TMS method [12]. The FF phase function is strongly forward-peaked, so without truncation N would have to be in the hundreds.
- **Interface.** At M1–M2 the interface is flat: Fresnel reflection and transmission are diagonal in the quadrature. Radiance in the refracted cone is mapped through Snell's law with the n^2 radiance factor, and total internal reflection applies beyond the critical angle. In the stage after M3, Cox–Munk rough-surface operators are integrated numerically per Fourier mode, for a wind speed w.
- **Solution.** In each mode, a homogeneous layer has the eigen solution of [14], with the reduced $N \times N$ eigenproblem. The direct beam adds a particular solution. The boundary conditions form a $2N \times 2N$ linear system:
  - at the top, downward radiance is the transmitted sky plus $\mathcal{R}_{wa}$ applied to the upward radiance;
  - at the bottom, upward radiance is ρ/π times the downward irradiance for $m = 0$, and zero for $m > 0$; for the emission case, a unit isotropic source.

  The sources share one factorisation of each mode's system.
- **Output.** Evaluate the upwelling radiance just below the surface at the view direction refracted into the water from the discrete-ordinate solution, using the source function integration of [14], then transmit it through the interface. Return R_0, E_{b0}, L_e and C for (4.2), for the direct and sky sources.

8.3 API

```
typedef struct {
  int n_streams;            // per hemisphere, default 24
  double n_water;           // 1.34
  double bp;                // FF backscatter fraction, default 0.0183
  double wind;              // 0: flat surface (M1-M2)
} rt_settings;

typedef struct {
  double R0, A, C;          // R_rs = R0 + A rho/(1 - C rho), per E_d(0+)
} rt_coeffs;

// Prepare moments, quadrature and interface operators; 0 on success.
int rt_init(const rt_settings *s, rt_workspace **w);
// Solve one node (omega, tau, f_w); geometry in air, degrees.
int rt_solve(rt_workspace *w, double omega, double tau, double f_w,
             double theta_sun, double theta_view, double dphi,
```

```
            rt_coeffs *direct, rt_coeffs *sky);
void rt_free(rt_workspace *w);
```

The optically deep limit, $\tau \to \infty$, is solved directly: there is no bottom boundary, and the upward radiance is the layer's reflection eigenvector solution. It gives R_0^∞ with $A = 0$.

8.4 Cost

The target is ≤ 1 ms per node on one core, with $N = 24$ and all modes. Most of the cost is the eigenproblems and the boundary-condition factorisation. They are shared by all three sources and by every view direction.

9 Tables (`rttable.c`)

9.1 Axes

Axis	Range	Transform	Nodes (default)
ω	0.02–0.995	$u = \ln(1-\omega)$	32
τ	10^{-3}–$\tau_{\max}$ (40), plus ∞	$v = \ln \tau$	48 + deep
f_w	0–1	linear	7

Geometry is fixed per table: one table per scene, for that scene's $(\theta_s, \theta_v, \Delta\phi)$ in air, each rounded to 0.1° for the cache key. Every node stores (R_0, A, C) for the direct and sky sources, 6 doubles, so a table is $32 \times 49 \times 7 \times 6 \times 8$ bytes ≈ 0.5 MB.

9.2 Build

Build all nodes with `rt_solve`; 11 000 nodes take about 10–20 s per scene at the target cost. Then validate (below). Tables are cached as NetCDF in `$PHOTIC_CACHE` (default `$HOME/.cache/photic/rt`), keyed by a hash of:

- the geometry;
- the `rt_settings`;
- the axes;
- the solver version string.

A scene whose table is in the cache costs nothing.

Cache file format. The file has the dimensions `omega`, `tau`, `fw`, `source` (2: direct, sky) and `coeff` (3: R0, A, C), and the variable `table(source, coeff, fw, tau, omega)` as a double. It holds the axis variables and the following global attributes:

- `theta_sun`, `theta_view` and `delta_phi`;
- `n_streams`, `n_water`, `bp` and `wind`;
- `solver_version`;
- the validation statistics: `max_rel_error`, `p95_rel_error` and `n_checks`.

9.3 Interpolation

Use tensor-product cubic Hermite interpolation in (u, v, f_w). The node derivatives come from centred finite differences of the node values in the transformed coordinates (one-sided at the ends). This makes the interpolant C^1 and exact for cubics. Interpolate the coefficients R_0, A and C, never R_{rs} itself, so (4.2) stays exact in ρ_b. Each evaluation returns the value and its three partial derivatives.

9.4 Validation of a table

After a build, solve 200 random off-node points directly, drawn uniformly in the transformed coordinates and $\rho_b \in [0, 1]$. Compare R_{rs} for $s = 0.2$. Record the maximum and 95th percentile relative errors in the file. Then:

- if the 95th percentile exceeds 0.3% or the maximum exceeds 1%, double the nodes along the axis with the largest error contribution, and rebuild;
- after two refinements, print a warning and continue.

The threshold is about a tenth of tier 4's noise level (R_{rs} noise about 5×10^{-4} on signals about 5×10^{-3}).

9.5 Derivatives for the fit

For one band sub-sample, with $D = H + \eta_t$,

$$\omega = b/c, \qquad \tau = cD, \qquad f_w = b_w/b, \qquad c = a + b, \qquad b = b_w + b_p.$$

The partial derivatives of the similarity variables with respect to the IOPs are

$$\frac{\partial \omega}{\partial a} = -\frac{b}{c^2}, \qquad \frac{\partial \omega}{\partial b_p} = \frac{a}{c^2}, \qquad \frac{\partial \tau}{\partial a} = \frac{\partial \tau}{\partial b_p} = D, \qquad \frac{\partial \tau}{\partial H} = c, \tag{9.1}$$
$$\frac{\partial f_w}{\partial b_p} = -\frac{b_w}{b^2}, \qquad \frac{\partial f_w}{\partial a} = 0.$$

The IOP derivatives with respect to $\ln P$, $\ln G$, $\ln X$ and Y are those of `water_iops`, with $b_p = b_{bp}/B_p$. For the bottom,

$$\frac{\partial R_{rs}}{\partial \rho_b} = \frac{A}{(1 - C\rho_b)^2}, \tag{9.2}$$

with $\partial \rho_b / \partial B_k$ and $\partial \rho_b / \partial w_k$ as in the existing `region_jacobian`, for cover weights w_k with $q_k = w_k / \sum w$. Band averaging (3.10) is linear, so the Jacobian rows are the same weighted sums of the sub-sample derivatives. All derivatives are those of the interpolant, so the Gauss–Newton model and the depth σ (`DEPTH_SIGMA`) are consistent with the misfit the fit evaluates.

10 Changes to `samodel.c`

1. **Forward model switch.** Add a global `forward_model`, set by `SET FORWARD LEE|RT`, default LEE. Declare it in `common.h` and define it in `common.c`. In RT mode:
   - `samodel()` gets or builds each scene's table before the pixel loop, and prints the table's validation statistics;

- `samodel_Rrs` computes each band from its sub-samples, (3.10) and (4.3), through the table;
- it fills `rrs_dp` with R_0^∞ and `rrs_bottom` with the bottom term, which the optically deep test and the index of optical depth use.

2. **Sub-samples.** For a scene with `SRF`, the sub-samples are 5 nm steps across the band's response, with weights $S_j F_0 \Delta\lambda$, precomputed per scene and band. The IOP tables are interpolated at each sub-sample. Without `SRF`, there is one sub-sample at the band wavelength.

3. **Jacobians.** `region_jacobian` and `water_jacobian` get RT branches built from §9.5. Until they exist, RT mode uses the existing finite-difference path (`ANALYTIC_JACOBIAN 0`), which is correct but slower.

4. **Removed in RT mode.** Lee's coefficients (`LEE_*`), the D_u path elongations, `INTERREFLECTION` (now exact in (4.2)) and the conversion $0.5\,r_{rs}/(1-1.5\,r_{rs})$ are all subsumed by the table.

5. **Kept.** The parameter vector, bounds, prior rows, `GLINT_OFFSET`, `Y_MODEL`, the bottom mixture, profile search, LUT warm starts, depth σ and outputs.

6. **The cache of IOPs.** `iop_cache` is extended with the sub-sample $(\omega, \tau/D, f_w)$ per scene, so that only τ is recomputed when H changes.

11 The atmosphere (`atmos.c`, M3)

11.1 Data

ACOLITE distributes 6SV-based LUTs per sensor, already averaged over each band's response (`ACOLITE-LUT-202110`). There is one NetCDF file per aerosol model (MOD1 continental, MOD2 maritime) and per pressure (500, 750, 1013 and 1100 hPa). Each band variable has the dimensions `(par, azi, thv, ths, wnd, tau)`. The `par` attribute lists the parameters, including `romix` (path reflectance), `utott` and `dtott` (total up and down transmittances), `astot` (spherical albedo) and the direct components. The axes are relative azimuth 0–180°, view and sun zenith 0–80°, and τ_{550} 0.001–5. They are not part of *photic*'s repository: ACOLITE's licence applies and they total 1.3 GB. They are read from the user's ACOLITE installation.

11.2 TOA model

$$\rho_j^{\mathrm{TOA}} = T_{g,j}\left[\rho_{\mathrm{path},j}(\tau_a) + \frac{T_j^{\downarrow}(\tau_a)\,T_j^{\uparrow}(\tau_a)\,\pi R_{rs,j}}{1 - S_j(\tau_a)\,\pi R_{rs,j}} + T_j^{\mathrm{dir}}\,\Delta_t\right]. \tag{11.1}$$

- ρ_{path}, $T^{\downarrow}$, $T^{\uparrow}$ and S are interpolated from the LUT: multilinear in angle and pressure, cubic in $\ln\tau_a$, with derivatives.
- T_g is the gas transmittance, from ACOLITE's gas LUT with the ozone and water vapour from the L1 metadata, as ACOLITE does.
- $R_{rs,j}$ is the band's water-leaving reflectance from the table, with the sky fraction $s = 1 - T^{\downarrow,\mathrm{dir}}/T^{\downarrow}$ in (4.3).
- Δ_t is the existing glint offset, now in TOA units.

11.3 In the fit

Add $\tau_{a,t}$ as a per-scene water-block parameter, bounded to $[0.001, 1.5]$. Its prior comes from a dark-spectrum estimate over the scene's deep and dark pixels in SWIR, which `atmos.c` computes once per scene, as ACOLITE's DSF does. The prior's σ is 0.03, or as set. The aerosol model is a discrete choice per scene, made in that first pass by the better SWIR fit, then fixed. The bands are the scene's visible, NIR and SWIR bands. Over water the NIR and SWIR water-leaving terms are taken as zero, which pins τ_a.

11.4 Inputs

Use L1 TOA reflectance grids, written by `validation/tier4_prepare.py` from ACOLITE's L1R output (`l1r_export_geotiff`). They are resampled to the site grid like the L2W bands. Scenes get the keyword `TOA` (§12).

12 Commands and data

12.1 Commands

Command	Effect
`SET FORWARD LEE|RT`	Forward model of `MODEL Lee_Semi_Analytical` (default LEE).
`SET RTSTREAMS n`	Streams per hemisphere (default 24).
`SET RTBP x`	Fournier–Forand backscatter fraction (default 0.0183).
`SET RTCACHE dir`	Table cache directory.
`COMPUTE RTTABLE SCENE s`	Build (or load) scene s's table and print its validation.
`PRINT RTTABLE SCENE s`	Print a table's settings and statistics.
`SCENE ... SRF f [AOT x]`	M2: τ_{550} for the sky fraction (default: from `ATMOSLUT` if given, else Rayleigh only).
`SCENE ... SRF f TOA ATMOSLUT dir`	M3: the bands are TOA reflectance; ACOLITE LUT directory `[PRESSURE hPa] [AEROSOL MOD1|MOD2|FIT] [OZONE cm] [WATERVAPOUR cm]`.

`SCENE` keywords stay positional and optional, after `SRF`, in the order shown. RT mode requires `SRF` on every scene. Without it, photic prints a warning and uses one sub-sample at the band wavelength.

12.2 New data files

- `model/data/f0_thuillier_2003.csv`: extraterrestrial solar irradiance, 1 nm, 350–1000 nm, from Thuillier et al. [19], for the weights in (3.10).
- `model/data/srf/`: add the NIR and SWIR bands for M3 (S2 B8, B8A, B11, B12; OLI B5, B6, B7).

- No RT tables are committed: they are built per scene and cached. The optional global table (Appendix A) would be a downloadable data file.

13 Tests

13.1 Unit and solver tests (`model/tests/rt_check.c`, `make test-rt`)

	Test	Pass
T1	Energy: $\omega = 1$, white bottom; flux in equals flux out.	Relative error $< 10^{-6}$.
T2	Thin layer: $\tau = 10^{-4}$, compared with analytic single scattering plus Fresnel reflection of the bottom.	$< 0.1\%$.
T3	Convergence in N: 16, 24, 32 and 48 streams.	R_{rs} changes by $< 0.2\%$ from 24 to 48.
T4	Against OSOAA (scalar build), on the tier 2 grid of water types, depths, bottoms and sun angles.	Median $< 0.5\%$, 95th percentile $< 1.5\%$, for $\omega \leq 0.95$.
T5	Closed form (4.2): direct solves at $\rho_b = 0, 0.3, 0.7, 1$ against the coefficients.	$< 10^{-6}$.
T6	Uniform-sky approximation: a uniform sky against a Rayleigh sky, from the atmosphere tables, for $\theta_s = 20$–$60°$.	Report; expected $< 1\%$ at near-nadir view.
T7	Table interpolation: the 200-point validation of every table.	95th percentile $< 0.3\%$, maximum $< 1\%$.
T8	Jacobian: interpolant derivatives against centred differences of the interpolant.	$< 10^{-5}$ relative.
T9	ACOLITE LUT reader: `romix`, `dtott` etc. at nodes, against direct reads of the NetCDF file.	Exact at nodes.
T10	TOA model: ACOLITE's own ρ_w for a scene, put through (11.1) at ACOLITE's τ_a, against the L1R TOA.	Agreement within ACOLITE's conventions; report the difference.
T11	Aerosol retrieval on synthetic TOA scenes (OSOAA with aerosol, PHOTIC_TIER3_SKY).	τ_a within 0.02.

`make test-rt` also runs `synth_gen` in RT mode: synthetic scenes from the RT forward model, inverted by `MODEL Lee_Semi_Analytical` with `SET FORWARD RT`. Its outputs are compared with a committed baseline, as `make test` does now. The existing `make test` is unchanged.

13.2 Benchmark

- **Tier 2.** RT mode against OSOAA on the existing cases. This is T4 at scale, and replaces the tier's role of measuring Lee's structural error.

- **Tier 3.** The `_insitu` scenarios, with bottoms from held-out measured spectra. They are the main synthetic test. A new water truth, OSOAA at 5 nm sub-samples and then band-averaged, makes tier 3 independent of the band-centre tables. It is built once and cached.
- **Tier 4.** Exmouth on Sentinel-2 and Landsat 8 (Tantabiddi, Bundegi, Marina, Gulf entrance); later, the Arabian Gulf site with ICESat-2 ATL24 truth. M3 runs from L1 TOA reflectance.
- **Acceptance.** Use the existing rule: tier 3 and tier 4 depth metrics must improve, with no regressions beyond the noise floor of about ±0.5 points or ±0.3 m RMSE at one site. Run time is reported. The target is at most 2× the present model once each scene's table is cached.

14 Risks and open questions

- **The bottom library.** An exact forward model cannot correct bottom shapes that do not match the seabed. In the `_insitu` tier 3 scenarios, fitting a single type per pixel against heterogeneous measured bottoms was much worse than fitting mixtures (RMSE 5–6 m against about 2 m in clear water), which is why the bottom is a mixture. The library's classes and their number remain open.
- **Aerosol–water trade-off.** With four visible bands, τ_a and X can trade off. The defences are the SWIR bands, the shared seabed across scenes and the prior from the dark spectrum.
- **Sky illumination.** The uniform sky approximation is quantified by T6. If it matters, add an atmosphere-derived sky radiance distribution as a third source.
- **Raman scattering and fluorescence** are not in the elastic RTE. Raman adds a few per cent in clear blue water. A correction table can be added later.
- **Stratified water** and non-Lambertian bottoms are outside the table.
- **Surface roughness** is flat in M1–M2. Cox–Munk operators come with the TOA model.
- **Licences.** ACOLITE's LUTs are read in place, not redistributed. The in situ spectra are CC BY 3.0 and their attribution is in the data file.

A A global table (optional)

The per-scene table can be replaced by one global table with the sun angle (12 nodes) and view direction (8 zenith nodes, with azimuthal Fourier coefficients) as further axes. That is about 10^5 solves, or minutes on a workstation, and a file of some tens to a few hundred megabytes. It is worth doing only if per-scene builds become a bottleneck, for example with many short scenes.

B Global solve with the adjoint (M6, optional)

When the atmosphere is a smooth field coupled across all pixels, or adjacency is modelled, the inversion becomes one PDE-constrained problem over the whole site:

$$\min_{u} \ \tfrac{1}{2} \sum_{t,j} \left\| \frac{\rho^{\mathrm{TOA}}_{t,j}[u] - \rho^{\mathrm{obs}}_{t,j}}{\sigma_{t,j}} \right\|^2 + \alpha_H \, \mathrm{TV}_\epsilon(H) + \alpha_w \sum_t \|\nabla \theta^t_w\|^2 + \alpha_a \sum_t \|\Delta \theta^t_a\|^2. \tag{B.1}$$

Write the RTE and its boundary conditions as $\mathcal{A}(u)L = q$. The adjoint radiance L^* solves $\mathcal{A}^* L^* = \sum_j \mathcal{M}_j^* \, (\rho_j^{\mathrm{TOA}} - \rho_j^{\mathrm{obs}})/\sigma_j^2$. The gradient is then $\partial \mathcal{J}/\partial u_i = -\langle L^*, \partial_{u_i} \mathcal{A}\, L \rangle$, whatever the number of unknowns [2, 8].

For depth, map the column to $s = z/D \in (-1, 0)$. D then becomes a coefficient, $\mu\, \partial_s L = D(-cL + \int \beta L)$, and the gradient is

$$\frac{\partial \mathcal{J}}{\partial H} = \sum_t \int_{-1}^{0} \int L_t^* \Big(- c L_t + \int \beta L_t \Big) \, \mathrm{d}\Omega \, \mathrm{d}s. \tag{B.2}$$

Matrix-free Gauss–Newton uses tangent-linear products for Jv and adjoint products for $J^{\mathsf{T}} w$. With the table forward model, only the atmospheric and adjacency couplings need adjoints; the column part is the table's own derivative.

References

[1] S. Blake, *photic*: a physics-based shallow water bathymetry model for multispectral imagery, arXiv:2002.02298 (2020).

[2] M. A. Box, S. A. W. Gerstl, C. Simmer, Application of the adjoint formulation to the calculation of atmospheric radiative effects, *Beitr. Phys. Atmos.* 61 (1988) 303–311.

[3] M. Chami, B. Lafrance, B. Fougnie, J. Chowdhary, T. Harmel, F. Waquet, OSOAA: a vector radiative transfer model of coupled atmosphere–ocean system for a rough sea surface, *Opt. Express* 23 (2015) 27829–27852.

[4] C. Cox, W. Munk, Measurement of the roughness of the sea surface from photographs of the sun's glitter, *J. Opt. Soc. Am.* 44 (1954) 838–850.

[5] K. F. Evans, The spherical harmonics discrete ordinate method for three-dimensional atmospheric radiative transfer, *J. Atmos. Sci.* 55 (1998) 429–446.

[6] G. R. Fournier, J. L. Forand, Analytic phase function for ocean water, *Proc. SPIE* 2258, Ocean Optics XII (1994) 194–201.

[7] M. Gao, P.-W. Zhai, B. A. Franz, et al., Inversion of multiangular polarimetric measurements over open and coastal ocean waters: a joint retrieval algorithm for aerosol and water-leaving radiance properties, *Atmos. Meas. Tech.* 12 (2019) 3921–3941.

[8] M. Hinze, R. Pinnau, M. Ulbrich, S. Ulbrich, *Optimization with PDE Constraints*, Springer (2009).

[9] Z. Jin, K. Stamnes, Radiative transfer in nonuniformly refracting layered media: atmosphere–ocean system, *Appl. Opt.* 33 (1994) 431–442.

[10] C. D. Mobley, *Light and Water: Radiative Transfer in Natural Waters*, Academic Press (1994).

[11] C. D. Mobley, L. K. Sundman, C. O. Davis, et al., Interpretation of hyperspectral remote-sensing imagery by spectrum matching and look-up tables, *Appl. Opt.* 44 (2005) 3576–3592.

[12] T. Nakajima, M. Tanaka, Algorithms for radiative intensity calculations in moderately thick atmospheres using a truncation approximation, *J. Quant. Spectrosc. Radiat. Transfer* 40 (1988) 51–69.

[13] C. D. Mobley (ed.), Ocean Optics Web Book, https://www.oceanopticsbook.info/view/introduction/overview.

[14] K. Stamnes, S.-C. Tsay, W. Wiscombe, K. Jayaweera, Numerically stable algorithm for discrete-ordinate-method radiative transfer in multiple scattering and emitting layered media, *Appl. Opt.* 27 (1988) 2502–2509.

[15] F. Steinmetz, P.-Y. Deschamps, D. Ramon, Atmospheric correction in presence of sun glint: application to MERIS, *Opt. Express* 19 (2011) 9783–9800.

[16] D. Tanré, M. Herman, P.-Y. Deschamps, Influence of the background contribution upon space measurements of ground reflectance, *Appl. Opt.* 20 (1981) 3676–3684.

[17] D. R. Thompson, E. J. Hochberg, G. P. Asner, et al., Airborne mapping of benthic reflectance spectra with Bayesian linear mixtures, *Remote Sens. Environ.* 200 (2017) 18–30.

[18] D. R. Thompson, V. Natraj, R. O. Green, et al., Optimal estimation for imaging spectrometer atmospheric correction, *Remote Sens. Environ.* 216 (2018) 355–373.

[19] G. Thuillier, M. Herse, D. Labs, et al., The solar spectral irradiance from 200 to 2400 nm as measured by the SOLSPEC spectrometer from the ATLAS and EURECA missions, *Solar Phys.* 214 (2003) 1–22.

[20] E. F. Vermote, D. Tanré, J.-L. Deuzé, M. Herman, J.-J. Morcrette, Second Simulation of the Satellite Signal in the Solar Spectrum, 6S: an overview, *IEEE Trans. Geosci. Remote Sens.* 35 (1997) 675–686.

[21] W. J. Wiscombe, The delta-M method: rapid yet accurate radiative flux calculations for strongly asymmetric phase functions, *J. Atmos. Sci.* 34 (1977) 1408–1422.